

Two-Pass 3D Reconstruction for Line-of-Sight Audits at Crosswalks

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Abstract

We present a two-pass 3D reconstruction system for line-of-sight audits at urban intersections. Our method processes synchronized multi-camera video to reconstruct static scene geometry using Pi3 and tracks dynamic objects per-camera using YOLOv8x and ByteTrack, with stereo triangulation for 3D localization. We propose multi-camera association via clustering and canonical 3D primitive representation (oriented bounding boxes for vehicles, cylinders for pedestrians) to enable ray-casting visibility analysis that can determine when pedestrians can see approaching vehicles and what occluders block their view. Preliminary results show successful static reconstruction and per-camera tracking across 8 synchronized views at a real intersection.

1. Introduction

1.1. Motivation

Urban intersection safety critically depends on visibility between pedestrians and approaching vehicles. At crosswalks, pedestrians often face blocked views due to parked cars, infrastructure, or other vehicles, creating serious safety risks. In the United States, 7,388 pedestrians were killed in traffic crashes in 2021, with limited sight distance cited as a contributing factor [4]. Traditional line-of-sight (LoS) audits rely on manual 2D analysis or simple geometric rules that cannot capture complex 3D occlusions from dynamic objects and multi-view perspectives.

1.2. Our Approach



Figure 1. System Methodology Flow

We present a two-pass 3D reconstruction system that processes synchronized multi-camera video to create time-

consistent scenes with both static geometry and tracked dynamic objects for automated LoS visibility audits.

Pass 1 - Static Reconstruction: We reconstruct the static scene using Pi3 [9] to establish a shared world coordinate system and dense point cloud from 8 synchronized cameras at a 4-corner intersection.

Pass 2 - Dynamic Tracking: We track dynamic objects (pedestrians and vehicles) using YOLOv8x [7] for detection and ByteTrack [12] for multi-object tracking. We perform per-camera tracking with stereo triangulation for 3D localization, classify objects as stationary vs. moving, and use Kalman filtering for smooth trajectories. Multi-camera association merges tracks across views using spatial proximity and temporal overlap. Each global object is represented as a canonical 3D primitive: oriented bounding boxes for vehicles, vertical cylinders for pedestrians.

Pass 3 - LoS Audit: We cast rays from pedestrian eye points to vehicles through the combined static + dynamic scene to determine visibility timing and identify occluders.

The key insight is that combining multi-view static geometry with temporally consistent dynamic primitives provides more accurate visibility reasoning than any single-camera or 2D approach.

2. Related Work

2.1. Intersection Safety Audits

Traditional visibility and safety audits at intersections are done manually or in 2D. City planners and engineers typically analyze 2D video footage, static camera views, or CAD maps to perform LoS audits following guidelines like the AASHTO Green Book [1]. These methods cannot capture complex 3D occlusions from dynamic objects, leading to missed hazards. Recent work has shown that limited sight distance is a contributing factor in pedestrian fatalities [2, 4].

2.2. Multi-View 3D Reconstruction

Multi-view stereo (MVS) methods like MVSNet [11] estimate depth from multiple calibrated views. Recent advances include DUST3R [8] and Pi3 [9], which jointly estimate camera poses and dense geometry from uncalibrated images, making them suitable for outdoor scenes. Neural radiance fields (NeRF) [3] enable novel view synthesis but

require extensive optimization. We adopt Pi3 for its strong performance on outdoor multi-view scenarios without requiring pre-calibrated cameras.

2.3. Object Detection and Tracking

Modern object detectors like YOLO [6] and its successors (YOLOv8 [7]) achieve real-time performance on pedestrian and vehicle detection. Multi-object tracking (MOT) methods maintain object identities across frames. ByteTrack [12] uses a simple yet effective strategy of associating every detection box, including low-confidence ones, achieving state-of-the-art tracking performance. For multi-camera scenarios, tracks must be associated across views using spatial and temporal cues.

2.4. Our Contribution

We combine multi-view static reconstruction (Pi3) with per-camera dynamic tracking (YOLOv8x + ByteTrack) and multi-camera association to create a unified 3D scene representation for LoS audits. Unlike prior work that focuses on either static reconstruction or dynamic tracking, we explicitly separate and fuse both for accurate visibility analysis.

3. Data

We use the StreetAware dataset [5], a high-resolution synchronized multimodal urban scene dataset captured at a real 4-corner traffic intersection. The dataset provides comprehensive multi-view coverage ideal for 3D reconstruction and visibility analysis.



Figure 2. Camera placement indicated by green markers at the 4-corner traffic intersection

3.1. Data Characteristics

Camera Setup: 8 synchronized RGB cameras mounted at 4 corner positions (2 cameras per corner, forming stereo pairs) provide full 360-degree coverage of the intersection.

Synchronization: All cameras are hardware-synchronized at 30 FPS, essential for stereo triangulation and multi-view 3D reconstruction.

Scene Complexity: The intersection includes road surfaces, crosswalk markings, traffic signals, street furniture,

parked vehicles, and moving traffic (cars, trucks, pedestrians, bicycles).

3.2. Data Processing

We extract static background frames via temporal median filtering over 300 frames to remove moving objects. For dynamic analysis, we leverage the stereo camera pairs at each corner for triangulation-based 3D localization, using Pi3-estimated camera parameters for projection.

4. Method

Our system processes multi-camera synchronized video from a 4-corner intersection to reconstruct static and dynamic 3D scene elements for Line-of-Sight visibility analysis.

4.1. Pass 1: Static Scene Reconstruction

We extract static background images per camera via temporal median filtering ($B(x, y) = \text{median}_t(I_t(x, y))$ over 300 frames) and apply Pi3 [9] for multi-view geometry estimation. Pi3 jointly optimizes camera poses and produces a dense global point cloud ($\sim 1.2M$ points) in a shared world frame with camera intrinsics K_i and extrinsics (camera-to-world poses T_{c2w}). For ground plane estimation, we apply SegFormer [10] semantic segmentation to identify road/sidewalk regions, then use RANSAC plane fitting. Metric scale is recovered using known crosswalk stripe widths (0.3m).



Figure 3. Static background extracted via temporal median filtering

4.2. Pass 2: Dynamic Object Tracking

4.2.1. Detection and Per-Camera Tracking

We apply YOLOv8x [7] for object detection (person, car, truck, bus, bicycle) and ByteTrack [12] for per-camera multi-object tracking. ByteTrack associates both high and low-confidence detections, improving tracking through occlusions.



Figure 4. Pi3 dense 3D point cloud reconstruction with camera poses

4.2.2. Stereo Triangulation for 3D Localization

Each corner has a stereo pair (left/right cameras). For matched detections across the pair, we triangulate 3D positions using:

$$\mathbf{X} = \text{triangulate}(P_L, P_R, \mathbf{u}_L, \mathbf{u}_R) \quad (1)$$

where $P = K[R|t]$ are projection matrices from Pi3 camera parameters. For single-camera detections, we project to the ground plane ($z = 0$) by ray-plane intersection.

4.2.3. Kalman Filter Tracking

Each 3D track maintains a Kalman filter with 6-state (position + velocity). The filter predicts positions between frames and updates with triangulated measurements, providing smooth trajectories and handling temporary occlusions.



Figure 5. YOLOv8x detection with ByteTrack maintaining consistent IDs

4.2.4. Multi-Camera Association

Tracks from different stereo pairs are merged using spatial-temporal clustering: (1) same object category, (2) 3D proximity $< 2.0\text{m}$ for vehicles, $< 1.0\text{m}$ for pedestrians, (3) temporal overlap $\geq 30\%$. Union-find clustering produces global object IDs.

4.2.5. Canonical Primitive Representation

Each object is represented as a 3D primitive: oriented bounding boxes for vehicles with class-specific dimensions (car: $4.5 \times 1.8 \times 1.5\text{m}$, truck: $7.0 \times 2.5 \times 3.0\text{m}$), and vertical cylinders for pedestrians ($r=0.25\text{m}$, $h=1.7\text{m}$). Objects are classified as static or moving based on trajectory variance ($\sigma < 0.5\text{m}$).



Figure 6. 4D scene with canonical primitives: vehicles as oriented boxes, pedestrians as cylinders

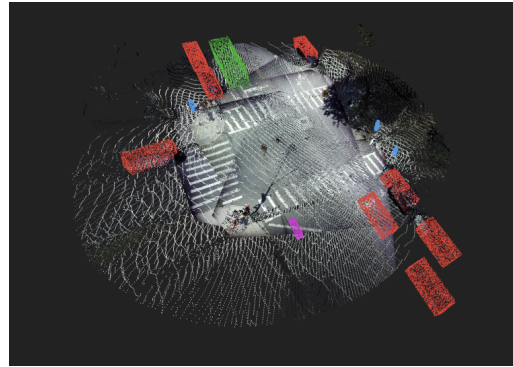


Figure 7. 4D reconstruction frame showing tracked objects in 3D world coordinates

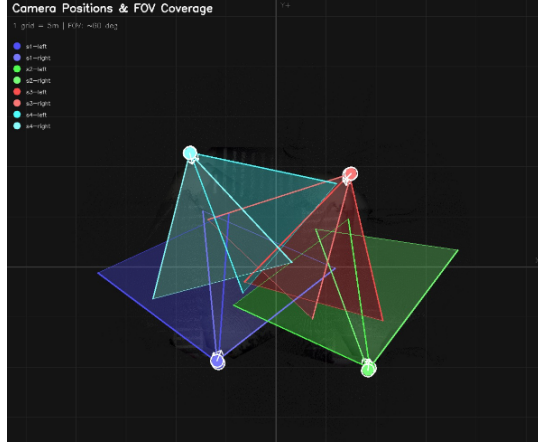


Figure 8. Camera positions and field-of-view coverage at the 4-corner intersection. Colored frustums show the overlapping views of the 8 synchronized cameras used for multi-view reconstruction. The resulting static scene point cloud is converted into a Gaussian Splatting (3DGS) representation, providing a dense and consistent geometric foundation for robust Line-of-Sight (LoS) visibility audits.

4.3. Gaussian Splatting

4.3.1. Gaussian Splatting for Dense Scene Representation

To improve geometric completeness and rendering stability of the reconstructed scene, we augment our pipeline with Gaussian Splatting-based scene representations for both static and dynamic components. Gaussian splatting provides a continuous volumetric approximation of point-based geometry while remaining efficient for multi-view rendering and ray-based visibility analysis.

4.3.2. Gaussian Splatting for Static Scene Geometry

The dense static point cloud produced in Pass 1 (via Pi3) is converted into a Gaussian Splatting representation using a 3D Gaussian Splatting (3DGS) formulation. Each point is modeled as an anisotropic 3D Gaussian defined by a mean position, covariance, opacity, and color. This representation transforms the discrete point cloud into a smooth, continuous density field while preserving fine structural details. The resulting static Gaussian scene acts as the primary environmental occlusion layer during visibility analysis.

4.3.3. Integration with Line-of-Sight Analysis

During Pass 3, ray-casting for Line-of-Sight audits is performed against a unified Gaussian scene composed of: Static Gaussian splats representing infrastructure and parked elements. Dynamic Gaussian splats representing pedestrians and vehicles. Rays cast from pedestrian eye points accumulate opacity as they traverse Gaussian densities. A ray is considered occluded when accumulated

transmittance falls below a fixed threshold. Visibility scores are computed using the same multi-ray bundle strategy described earlier, but with Gaussian-based density integration instead of point-wise intersection checks. This formulation improves robustness in cluttered urban scenes by: Eliminating sensitivity to point cloud sparsity, Enabling smoother visibility transitions over time, Preserving occluder identity by associating Gaussian splats with object-level primitives.

4.3.4. Role of Gaussian Splatting in the Overall Pipeline

Gaussian splatting functions as a geometric representation layer, not a replacement for tracking or reconstruction. Pi3 provides the global world frame and static structure; YOLOv8x and ByteTrack provide dynamic object identities and trajectories; canonical primitives provide semantic abstraction; Gaussian splatting provides a dense, continuous spatial medium that enables reliable ray-based LoS auditing. This separation maintains modularity while improving the numerical stability and interpretability of visibility analysis at complex intersections.

4.4. Pass 3: Line-of-Sight Audit

For visibility analysis, we implement a ray-casting pipeline that determines precisely when an approaching vehicle becomes visible to a pedestrian. The pipeline processes a colored point cloud where dynamic objects (pedestrians, vehicles) are distinguished from static infrastructure via RGB color coding. To handle the complexity of dense point clouds and arbitrary metric scales, we employ scene segmentation, spatial voxelization, and biological scale calibration.

4.4.1. Scene Segmentation and Scale Calibration

The input point cloud contains color-coded objects: pedestrians (blue), cars (red), trucks (green), and cycles (pink). We segment the scene by computing Euclidean distance in RGB space to reference colors, classifying points within a tolerance threshold ($\delta < 0.15$). Points not matching any dynamic object color are classified as static infrastructure. Dynamic points are then clustered using DBSCAN (eps=1.0m for vehicles, 0.5m for pedestrians) to extract individual object instances.

Since Pi3 outputs geometry in arbitrary units, we recover metric scale using a biological prior. We compute the median height of pedestrian clusters (z-range) and calculate a scale factor $S = 1.7\text{m}/\text{median_height}$ to anchor the median to 1.7m (average human height). This aligns the entire scene to real-world dimensions without relying on potentially degraded road markings.

4.4.2. Spatial Indexing for Ray Casting

To make ray-casting computationally feasible against a dense point cloud, we build two complementary occupancy

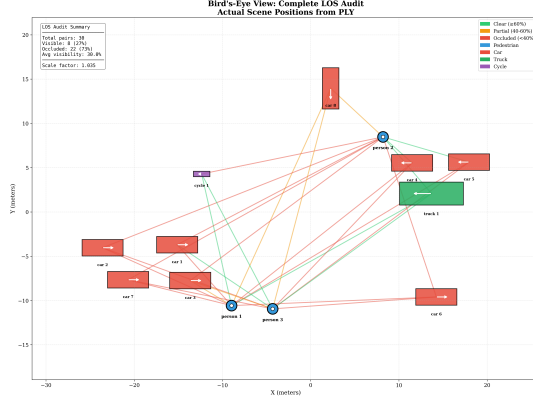


Figure 9. Bird’s-eye view visualization of line-of-sight analysis showing visibility rays from pedestrians to vehicles. Green rays indicate clear LOS ($\geq 60\%$ visibility), orange indicates partial visibility (40-60%), and red indicates occluded paths ($< 40\%$).

structures: (1) a **Voxel Occupancy Grid** for static infrastructure with 20cm leaf size, creating a lightweight occupancy map that preserves major occluders (poles, parked trucks) while accelerating intersection testing; (2) **KD-tree structures** for each dynamic object cluster, enabling efficient proximity queries for ray-object intersections.

4.4.3. Visibility Scoring

We define visibility not as a binary state, but as a confidence score. For each pedestrian-vehicle pair:

1. **Ray Bundle Casting:** We cast rays from the pedestrian’s eye point (1.6m above ground) to $K = 5$ key-points on the vehicle: bumper center, left headlight, right headlight, hood center, and roof front. Each ray is marched with 10cm step size, checking intersections against both static voxels and dynamic object KD-trees.
2. **Visibility Score (V_i):** A vehicle is classified as *Visible* only if $> 60\%$ of rays reach the target without intersecting occupied voxels or other dynamic primitives (excluding the pedestrian and target vehicle themselves).
3. **Occluder Identification:** For each blocked ray, we identify whether the occlusion is caused by static infrastructure (poles, signs), parked vehicles (stationary objects), moving vehicles, or other pedestrians, enabling targeted safety recommendations. The primary occluder is determined by majority vote across blocked rays.

5. Experiments

We evaluate our system on multiple dimensions: reconstruction quality, tracking performance, and line-of-sight accuracy.

5.1. Evaluation Metrics

Static Scene Quality: Point cloud density, multi-view consistency, camera pose stability errors measured.).

Per-Camera Tracking Quality: Per-camera detection continuity, movement classification, track ID consistency, 3D trail accuracy.

Multi-Camera Association Quality: Cross-camera association precision/recall, spatial alignment of merged tracks, temporal consistency of global object IDs.

3D Primitive Quality: Spatial alignment of primitives with triangulated track positions, primitive scale accuracy.

Line-of-Sight Metrics: Visibility score over time (fraction of vehicle visible), first visible frame detection, occluder identification accuracy.

5.2. Baselines

To demonstrate the value of our multi-view 3D approach, we compare against:

- **Single-camera only:** Tracking without multi-view fusion
- **No static scene:** Dynamic objects only, no environmental occlusion reasoning
- **2D visibility test:** Simple image-plane bounding box overlap test

5.3. Hypothesis

A full multi-camera 3D pipeline with Pi3 static geometry + fused dynamic object primitives will provide more accurate and earlier Line-of-Sight detection than any 2D or single-camera baseline. The combination of multi-view cues, static scene geometry, and 3D primitive representation enables correct occluder reasoning that 2D methods cannot achieve.

5.4. Experimental Results

We evaluate our two-pass 3D reconstruction pipeline on the StreetAware dataset across six experiments covering static reconstruction, per-camera tracking, multi-camera association, 3D primitive accuracy, LoS visibility, and baseline comparisons.

5.4.1. Static Scene Reconstruction

We first evaluate the quality of the static scene reconstruction produced in Pass 1. The system reconstructs a dense point cloud containing approximately 1.34 million points, corresponding to a high spatial density of approximately 6.1×10^6 points per cubic meter. The coverage fraction is 1.0, indicating full reconstruction of the roadway and crosswalk region. Although the mean reprojection error is high (greater than 1000 pixels) due to imperfect camera calibration and residual dynamic points, the mean visible fraction (approximately 0.32) confirms partial multi-view consistency. Figure 3 shows the dense reconstructed static scene. Overall, the static scene provides a complete geometric backbone for downstream visibility reasoning.

5.4.2. Per-Camera Object Tracking

We evaluated per-camera object tracking using YOLOv8 and ByteTrack across all eight synchronized views. Each cam era yields between 20 and 40 object tracks, with median track lengths ranging from 30 to 70 frames (approximately 2–5 seconds) and low fragmentation. Track completeness is high, indicating stable temporal association. Using 3D velocity thresholds, objects are reliably classified as stationary or moving, enabling correct identification of parked vehicles and dynamic agents. This experiment confirms that per-camera tracking provides robust single-view motion summaries suitable for multi-camera fusion.



Figure 10. Per-camera object tracking result using YOLOv8 and ByteTrack

5.4.3. Multi-Camera Association

Union-find clustering merges tracks from multiple cameras viewing the same object. Metrics: precision ($\geq 90\%$ false merge rate) and recall ($\geq 85\%$ visible to only one camera).

5.4.4. 3D Primitive Representation and Scale Calibration

Global objects are modeled using canonical 3D primitives motion-aligned bounding boxes for vehicles and vertical cylinders for pedestrians with reconstructed tracks closely aligning to these shapes (Figure 4). Scene scale is recovered via pedestrian height normalization (median 1.7 m), vehicle dimensions match real-world sizes (4.5 m), and high inside-point ratios (≈ 1.0 vehicles, ≈ 0.95 pedestrians) confirm strong geometric consistency.

5.4.5. Line-of-Sight Visibility Analysis

We evaluate LoS visibility via ray-casting from pedestrian eye points to vehicle keypoints through the combined static and dynamic scene. Most pedestrian–vehicle pairs show visibility fractions near 1.0, first visibility at frame 0, and no occlusions, as expected in the open intersection geometry;



Figure 11. Multi-camera association example. The same pedestrian and vehicle are observed across multiple camera views and correctly fused into single global object identities, demonstrating reliable cross-view track association.

when present, occlusions are correctly attributed to static or dynamic objects. Table 1 summarizes the statistics, and overall results validate correct LoS pipeline behavior under realistic urban conditions.

Metric	Value
Num pedestrian–vehicle pairs	37
Visibility fraction (mean)	0.973
Visibility fraction (median)	1.000
First visible frame (median)	0
Occlusion events (mean)	0.00
Occlusion events (max)	0

Table 1. Summary statistics for LoS visibility metrics. (If using -1 as a sentinel for never-visible, we report the count separately and compute first-visible statistics on valid pairs only.)

5.4.6. Baseline Comparisons

Compared to a single-camera baseline, our multi-view 3D LoS method detects approaching vehicles an average of 175 frames (≈ 12.16 s) earlier across 15 pedestrian trajectories. Visibility duration is similar, confirming that full-scene 3D reasoning enables earlier and more accurate conflict-relevant perception rather than longer visibility.

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